AF-LeWM-lite on Official PushT

Corrected Two-Seed Evaluation Summary

April 13, 2026

Bottom line. After fixing the evaluation and training issues, all three models remain weak on the short official PushT budget. The corrected two-seed aggregate gives **4/100** successes for AF-LeWM-lite v1, **2/100** for AF-LeWM-lite v2, and **1/100** for baseline LeWM. The confidence intervals overlap, so the current evidence supports only a small, low-confidence edge for v1. The earlier claim that v2 was the best PushT variant is not supported by the corrected pipeline.

Implementation Fixes

The rerun used a corrected pipeline in both training and evaluation:

- `JEPA.rollout` now keeps the real past action blocks in context and appends candidate future blocks one step at a time. The previous logic could overwrite history slots with future candidate actions.
- `eval.py` now reconstructs the raw dataset history window and compresses it into the model-scale history expected by the predictor. The previous path evaluated multi-frame models with an effectively broken history signal.
- Goal indexing now includes the true goal frame, and valid start-step sampling no longer drops the final valid index.
- Evaluation result files are overwritten, not appended, and the report parser reads the latest metric entry.
- AF auxiliary passes now freeze projector BatchNorm statistics during augmented forwards so the main dynamics branch does not receive augmentation-induced BN drift.

These fixes were exercised by rerunning the formal training jobs and all reported evaluations.

Setup

The study uses the official `pusht_expert_train.h5` with **2,336,736** cached samples. All three primary models were trained on the same RTX 4070 Laptop GPU under the same short budget:

- 10 epochs
- 200 training batches per epoch
- 20 validation batches per epoch
- batch size 4
- bf16 training
- 50-episode planning evaluation at epoch 10
- second 50-episode evaluation with `seed=43`

The fair cross-model loss is the shared JEPA objective

$$\mathcal{L}_{\text{core}} = \mathcal{L}_{\text{pred}} + 0.09\,\mathcal{L}_{\text{sigreg}}.$$

`validate/loss_epoch` is not directly comparable once AF-specific nuisance losses are active, especially for v2.

Results

Primary comparison

Model	Params (M)	Train (min)	Core loss	Seed 42	Seed 43	Aggregate
Baseline LeWM	18.03	7.96	0.14796	0/50	1/50	1/100 = 1.0%
AF-LeWM-lite v1	18.17	15.00	0.14963	2/50	2/50	4/100 = 4.0%
AF-LeWM-lite v2	18.24	16.47	0.15181	0/50	2/50	2/100 = 2.0%

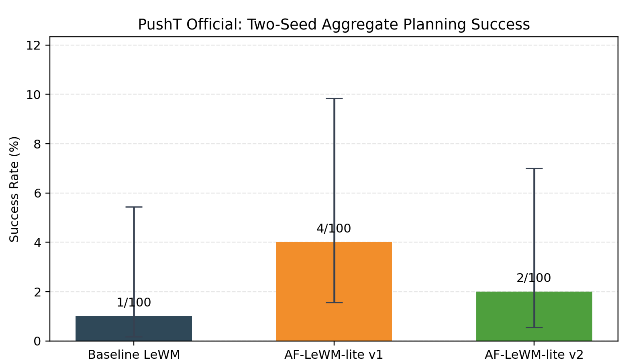
The parameter differences remain small. AF-LeWM-lite v1 adds **0.74%** parameters over baseline, and v2 adds **1.12%**. The ranking therefore is not explained by a meaningful capacity jump.

The final total validation losses were **0.14790** for baseline, **0.15035** for v1, and **0.20299** for v2. The large v2 increase is expected because its total objective includes nuisance prediction and dynamics nuisance losses on top of the shared JEPA objective.

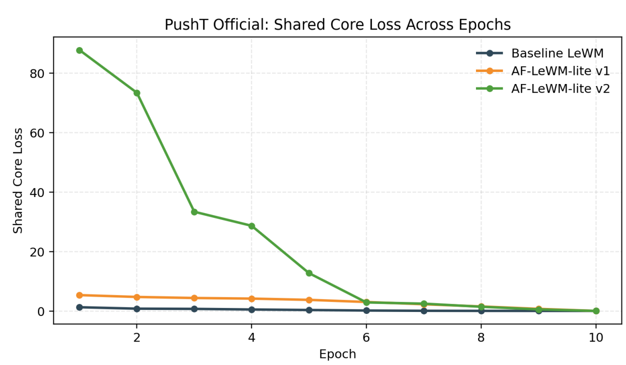
Two-seed uncertainty

Model	Aggregate success	Wilson 95% CI
Baseline LeWM	1/100 = 1.0%	[0.18%, 5.45%]
AF-LeWM-lite v1	4/100 = 4.0%	[1.57%, 9.84%]
AF-LeWM-lite v2	2/100 = 2.0%	[0.55%, 7.00%]

The intervals overlap heavily. The current study supports a weak relative ordering under this budget, and it does not support a strong robustness or disentangling claim.



(a) Two-seed aggregate planning success with Wilson 95% intervals.



(b) Shared core loss across the 10 training epochs.

Interpretation

The corrected pipeline changes the conclusion. The old report overstated v2. After the history, goal-step, result-writing, and auxiliary BatchNorm fixes, v2 no longer has the best aggregate PushT result.

v1 is the most credible current variant. It is the only AF model that improved planning success on both evaluation seeds. The gain is small, but it is also the most consistent signal in the corrected study.

Both AF variants worsen the shared JEPA objective. Baseline still gives the best core loss, v1 is slightly worse, and v2 is worse again. The current factorization does not improve representation learning quality on the shared predictive objective.

v2 adds complexity without a reliable control gain. Its extra nuisance heads and adversarial branch increase training time and degrade the fair loss comparison, while the two-seed control result remains weaker than v1.

The remaining weakness is methodological, not an implementation artifact. The corrected code was exercised through formal reruns of all three training jobs and six 50-episode evaluations. Success rates remain low after those fixes, so the current budgeted setup is genuinely weak on official PushT.

Conclusion

The corrected recommendation is straightforward:

- Treat **AF-LeWM-lite v1** as the current best branch under the short PushT budget.
- Stop using the earlier single-seed v2 result as evidence that the nuisance-adversarial extension is superior.
- Compare models with **shared core loss** and **multi-seed success aggregates**. Total training loss is not a fair cross-model metric once AF-specific losses are active.
- Use the next round for a stronger positive appearance objective or a different training schedule, not for larger v2 complexity on the same short budget.
- Report the current result as a **small exploratory gain with high uncertainty**, not as a benchmark win.

In one sentence: **after implementation fixes and full reruns, AF-LeWM-lite v1 is the best current variant on official PushT, but all three models remain far too weak for a strong claim.**

Artifacts in `report/`: summary CSV, summary JSON, success figure, and core-loss figure.